Qidong He — Curriculum Vitae

43 Leon St – Boston, MA 02115 – USA
q.he@northeastern.edu • qidong-he.github.io

Academic Positions

Northeastern University **Boston, MA**
Zelevinsky postdoctoral fellow *2026–2029*
Mentor: Matan Harel

Education

Rutgers University–New Brunswick **Piscataway, NJ**
Ph.D., mathematics *2021–2026*
Thesis: Phase transitions in hard-core models of statistical mechanics
Advisors: Ian Jauslin & Joel Lebowitz

Rutgers University–New Brunswick **Piscataway, NJ**
M.S., mathematics *2021–2023*

Colby College **Waterville, ME**
B.A., summa cum laude, mathematics & physics with a minor in Japanese *2017–2021*
Thesis: Counting conjugacy classes of elements of finite order in compact exceptional groups

Research Interests

Packing problems, statistical physics, and probability theory, with an emphasis on phase transitions in lattice models of solids and liquid crystals

Research Papers

Preprints

[12] Alexander Barg, Qidong He, and Geyang Wang. A general framework for crystallization in maximal hard-core models. 2026. arXiv: `2610.11042 [cond-mat.stat-mech]`.

[11] Qidong He. Nematic order in monomer-dimer models of Heilmann and Lieb with lateral attraction in dimensions $d \geq 2$. 2026. arXiv: `2610.08600 [math-ph]`.

[10] Qidong He. A new upper bound on the dimer constant of $\mathbb{Z}^3$. 2026. arXiv: `2607.28810 [math.CO]`.

[9] Qidong He. Rigorous bound on the aspect ratio for the formation of a nematic phase in hard rod and hard rectangle systems on $\mathbb{Z}^2$. 2026. arXiv: `2607.10510 [math-ph]`.

[8] Qidong He. Unified criteria for crystallization in hard-core lattice systems with applications to polyomino fluids and chiral mixtures. 2026. arXiv: `2602.05294 [math-ph]`.

[7] Qidong He. Extended regime of nematic order in an interacting monomer-dimer model of Heilmann and Lieb. 2025. arXiv: 2511.23437v2 [math-ph].

Publications

[6] Qidong He, Ian Jauslin, Joel Lebowitz, and Ron Peled. Liquid-vapor transition in a model of a continuum particle system with finite-range modified Kac pair potential. *Communications in Mathematical Physics*, 407(11):240, 2026. DOI: 10.1007/s00220-026-05735-w.

[5] Qidong He. Upper bounds for the connective constant of weighted self-avoiding walks. *Journal of Physics A: Mathematical and Theoretical*, 58(50):5003, 2025. DOI: 10.1088/1751-8121/ae280e.

[4] Qidong He. Analyticity for locally stable hard-core gases via recursion. *Journal of Statistical Physics*, 192(4):46, 2025. DOI: 10.1007/s10955-025-03435-8.

[3] Qidong He and Ian Jauslin. High-fugacity expansion and crystallization in non-sliding hard-core lattice particle models without a tiling constraint. *Journal of Statistical Physics*, 191(10):135, 2024. DOI: 10.1007/s10955-024-03349-x.

[2] Tamar Friedmann and Qidong He. Counting conjugacy classes of elements of finite order in exceptional Lie groups. *Combinatorial Theory*, 4(1), 2024. DOI: 10.5070/C64163850. Undergraduate research.

[1] Qidong He and Scott A Taylor. Links, bridge number, and width trees. *Journal of the Mathematical Society of Japan*, 75(1):73–111, 2023. DOI: 10.2969/jmsj/86158615. Undergraduate research.

Talks

Applied and Interdisciplinary Mathematics (AIM) Seminar: Nematic liquid-crystalline phase in an interacting monomer-dimer model of Heilmann and Lieb. Northeastern University, *Sep 2026*

130th Statistical Mechanics Conference: High-density crystallization of polyomino fluids on the square lattice (short talk). Rutgers University, *May 2026*

Rutgers Graduate Student Combinatorics Seminar: Crystalline order in random packings of Z-pentominoes on the square lattice. Rutgers University, *Mar 2026*

129th Statistical Mechanics Conference: Extended regime of nematic order in an interacting monomer-dimer model of Heilmann and Lieb (short talk). Rutgers University, *Dec 2025*

Rutgers Graduate Student Pizza Seminar: Structures of random packings. Rutgers University, *Oct 2025*

Rutgers Graduate Student Combinatorics Seminar: Random packings. Rutgers University, *Apr 2025*

126th Statistical Mechanics Conference: Analyticity for classical hard-core gases via recursion (short talk). Rutgers University, *May 2024*

124th Statistical Mechanics Conference: Crystallization phenomenon in a large class of hard-core lattice particle models (short talk). Rutgers University, *May 2023*

Colby Liberal Arts Symposium (CLAS): Counting conjugacy classes of elements of finite order in G_2. Colby College, *Apr 2021*

North Shore Undergraduate Math Conference: Knot Invariants. Gordon College, *Apr 2019*

Mathematics and Statistics Colloquium: Knot Invariants. Colby College, *Sep 2018*

Teaching Experience

TA, Department of Mathematics, Rutgers University–New Brunswick:
*Recipient of the TA Teaching Excellence Award**
- Recitation instructor: multivariable calculus (F23*, F22)
- TA-at-large: advanced calculus for engineering (F24, Sp24), introductory linear algebra (Sp25, Sp23)
- Grader: advanced calculus for engineering (Sp22), cryptography (Sp22), introduction to applied mathematics (F21), linear algebra (F21)

TA, Department of Mathematics and Statistics, Colby College:
- Grader: abstract algebra (F20), combinatorics (Sp21), honors calculus I (F19), real analysis (Sp20), single-variable calculus (Sp19, F18)

TA, Department of Physics and Astronomy, Colby College:
- Grader: foundations of electromagnetism and optics (Sp20)

Service

Mentoring

Rutgers Directed Reading Program (DRP):
- Aditya Agarwal (F25)
- Keshav Badri, Phase transitions in the Ising model (Su25)
- Derek Chan, Calculating effective volume for a Gaunt–Fisher configuration (Sp25)
- Harry Haedrich, Non-sliding hard-core lattice particle models—proving crystallization (Sp25)

Refereeing

- Journal of Statistical Physics (1x)
- Combinatorics, Probability and Computing (1x)

Honors and Awards

SGS Research & Conference Travel Award: Rutgers University–N.B. ($1,000) *Mar 2026*

SAS Fellowship: Rutgers University–New Brunswick ($30,833) *Aug 2025*

SGS Research & Conference Travel Award: Rutgers University–N.B. ($1,000) *Mar 2025*

TA Teaching Excellence Award: Rutgers University–New Brunswick *Feb 2024*

Academic Excellence Award: Rutgers University–New Brunswick *Jan 2022*

Marston Morse Prize in Mathematics: Colby College *May 2021*

Julius Seelye Bixler Scholar: Colby College *Sep 2020, Sep 2019*

William A. Rogers Prize in Physics and Astronomy: Colby College *May 2020*
Helen and Robert E. L. Strider Scholar: Colby College *Sep 2018*

Memberships

Sigma Pi Sigma: *May 2021*
Phi Beta Kappa: *Apr 2020*